SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0607 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED
CO2:*NeuralNetworkRepresentation, Perceptron Model, Multilayer Perceptrons,*

Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

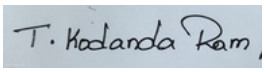
Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:
IT . KODANDA RAM (192524162) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
Iunderstand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6.

For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

9. A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

14. A Genetic Algorithm is applied to maximize the function f(x) = 3x + 2 for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima. (5 Marks)

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

ITA0607 - MACHINE LEARNING

ASSESSMENTTOOL 1:ANALYTICALPROBLEMSOLVING

Worked answers - 15 questions

The solutions below use the values given in the question paper. Each numerical answer shows the formula, substitution, working, and result.

Working convention

For perceptron questions, the step output is 1 when z is greater than or equal to 0; otherwise it is 0.

For the weight-update questions, the delta rule used is Delta w = eta(t - y)x, and the updated weight is w_new = w_old + Delta w.

Percentages are rounded to two decimal places. For genetic-algorithm questions, the binary encoding and crossover point are stated where the paper does not give them.

1. Perceptron - Student pass or fail

Question: A simple perceptron model isu sed toclassify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

Given: x1 = 2, x2 = 1, w1 = 0.6, w2 = 0.4, b = -0.5.

First, calculate the net input.

Formula: z = w1x1 + w2x2 + b

Substitute: z = (0.6 x 2) + (0.4 x 1) + (-0.5)

z = 1.2 + 0.4 - 0.5 = 1.1

Now apply the step function: f(z) = 1 when z >= 0; otherwise f(z) = 0. Since 1.1 is greater than 0, the output is 1.

Answer: The net input is 1.1. The output is 1, so the student is classified as Pass.

2. Weight update using the delta rule

Question: In a neuraln etwork, a single neuronis trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Given: w_old = 0.5, eta = 0.1, x = 2, y = 0.6, t = 1.

The output error is target minus actual output.

Formula: e = t - y

Substitute: e = 1 - 0.6 = 0.4

Next, calculate the weight change.

Formula: Delta w = eta(t - y)x

Substitute: Delta w = 0.1(1 - 0.6)(2) = 0.08

Add this change to the original weight.

Formula: w_new = w_old + Delta w

w_new = 0.5 + 0.08 = 0.58

Answer: Error = 0.4, weight change = 0.08, and the new weight is 0.58.

3. Evaluation of a disease prediction model

Question: A deep learning model is developedf ordisease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Given: TP = 180, TN = 150, FP = 20, FN = 50.

Total number of cases: N = TP + TN + FP + FN = 180 + 150 + 20 + 50 = 400.

Accuracy

Formula: Accuracy = (TP + TN)/N x 100

Substitute: Accuracy = (180 + 150)/400 x 100 = 82.50%

Precision

Formula: Precision = TP/(TP + FP) x 100

Substitute: Precision = 180/(180 + 20) x 100 = 90.00%

Sensitivity (recall)

Formula: Recall = TP/(TP + FN) x 100

Substitute: Recall = 180/(180 + 50) x 100 = 78.26%

Specificity

Formula: Specificity = TN/(TN + FP) x 100

Substitute: Specificity = 150/(150 + 20) x 100 = 88.24%

Metric	Result
Accuracy	82.50%
Precision	90.00%
Sensitivity / Recall	78.26%
Specificity	88.24%

Discussion: Precision is high, but the model misses 50 actual disease cases and has a recall of 78.26%. That is a concern in medical diagnosis. It could be considered for preliminary screening, but these results alone are not enough to establish that it is reliable for independent diagnosis. Sensitivity and the consequences of missed cases need careful attention.

Answer: Accuracy = 82.50%, precision = 90.00%, recall = 78.26%, and specificity = 88.24%.

4. Genetic Algorithm - Maximizing f(x) = x^2

Question: A Genetic Algorithmi s applied tomaximize thefunction f(x) = x^2 using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Given: population = {2, 4, 6, 8}; fitness function f(x) = x^2.

Calculate the fitness of each individual using f(x) = x^2.

Individual	Calculation	Fitness
2	2^2	4
4	4^2	16

Individual	Calculation	Fitness
6	6^2	36
8	8^2	64

The two highest fitness values are 64 and 36, so the selected candidates are 8 and 6.

Using four-bit binary, 8 = 1000 and 6 = 0110. The least significant bit is the rightmost bit.

Exchange the least significant bits: 100l0 and 011l0. Both bits are 0, so the offspring do not change.

Offspring 1 = 1000 = 8; Offspring 2 = 0110 = 6

Mutation randomly changes a bit. For example, 1000 can become 1001, changing the value from 8 to 9. Mutation adds variety to the population and helps the algorithm explore solutions that crossover alone may not produce.

Answer: Fitness values are 4, 16, 36, and 64. The best candidates are 8 and 6. The LSB exchange gives the same offspring because both last bits are zero.

5. Training accuracy and validation accuracy

Question: Duringt he trainingof a deepneural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

The training accuracy keeps increasing, but validation accuracy falls from 72% to 68%. This indicates overfitting.

Overfitting happens when a network learns the training examples too closely, including noise or details that do not help with unseen data. It can happen when the model is too complex, training continues for too long, or regularization is insufficient.

Technique 1: Early stopping. Check validation performance after each epoch and stop training when it stops improving.

Technique 2: Dropout. Randomly deactivate some neurons during training so the network does not depend too heavily on a small set of neurons.

Validation accuracy is useful because it shows how the model performs on data that is not used to update its weights.

It helps identify overfitting and choose when to stop training.

Answer: The model is overfitting. Early stopping and dropout are two ways to reduce this problem.

6. Perceptron model - Loan approval

Question:Aperceptron model is usedto classify whether a loan is approved based on two features: income and credit score.The weights are 0.7 and 0.5, and the bias is -0.6. For an applicant with income = 1.5 and credit score = 2, compute thenet input and apply the step activation function. Determine whether the loan is approved or rejected.

Given: x1 = 1.5, x2 = 2, w1 = 0.7, w2 = 0.5, b = -0.6.

Formula for net input: z = w1x1 + w2x2 + b

Substitute: z = (0.7 x 1.5) + (0.5 x 2) - 0.6

z = 1.05 + 1.00 - 0.60 = 1.45

The step function gives 1 when z >= 0 and 0 otherwise. Since 1.45 >= 0, the output is 1.

Answer: Net input = 1.45; output = 1. The perceptron classifies the loan as Approved.

7. Weight update using the delta learning rule

Question: In a neural network, as ingle neuron uses thedeltalearning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration.

Given: w_old = 0.4, eta = 0.2, x = 3, y = 0.5, t = 0.9.

Formula for error: e = t - y

Substitute: e = 0.9 - 0.5 = 0.4

Formula for weight change: Delta w = eta(t - y)x

Substitute: Delta w = 0.2(0.9 - 0.5)(3) = 0.24

Formula for new weight: w_new = w_old + Delta w

Substitute: w_new = 0.4 + 0.24 = 0.64

Answer: Error = 0.4, weight change = 0.24, and new weight = 0.64.

8. Classification model evaluation

Question: A classification model produces thef ollowing confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical.

Given: TP = 120, TN = 200, FP = 30, FN = 50. Total N = 120 + 200 + 30 + 50 = 400.

Accuracy

Formula: Accuracy = (TP + TN)/N x 100

Substitute: Accuracy = (120 + 200)/400 x 100 = 80.00%

Precision

Formula: Precision = TP/(TP + FP) x 100

Substitute: Precision = 120/(120 + 30) x 100 = 80.00%

Recall

Formula: Recall = TP/(TP + FN) x 100

Substitute: Recall = 120/(120 + 50) x 100 = 70.59%

Specificity

Formula: Specificity = TN/(TN + FP) x 100

Substitute: Specificity = 200/(200 + 30) x 100 = 86.96%

F1-score

Formula: F1 = 2TP/(2TP + FP + FN)

Substitute: F1 = 240/(240 + 30 + 50) = 0.75 = 75.00%

Metric	Result
Accuracy	80.00%
Precision	80.00%
Recall / Sensitivity	70.59%
Specificity F1-	86.96%
score	75.00%

The false-negative rate is FN/(TP + FN) x 100 = 50/170 x 100 = 29.41%. If missing a positive case is serious, this rate is a concern. The model would need better recall before being used in such an application.

Answer: Accuracy = 80.00%, precision = 80.00%, recall = 70.59%, specificity = 86.96%, and F1-score = 75.00%.

9. Genetic Algorithm - Maximizing f(x) = 2x + 3

Question:A GeneticAlgorithm isu sedto maximizethe function f(x)= 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence.

Given: population = {1, 3, 5, 7}; fitness function f(x) = 2x + 3.

Use f(x) = 2x + 3 to calculate each fitness value.

Individual	Calculation	Fitness
1	2(1) + 3 2(3)	5
3	+ 3 2(5) + 3	9
5	2(7) + 3	13
7		17

The top two candidates are 7 and 5.

Using four-bit binary: 7 = 0111 and 5 = 0101. Choose a crossover point after the second bit.

Parents: 01|11 and 01|01. Exchange the parts after the point.

Offspring 1 = 01|01 = 0101 = 5; Offspring 2 = 01|11 = 0111 = 7

The offspring happen to be the same as the parents because the first two bits match. Mutation can change a bit, such as 0111 to 0110, introducing a new candidate and helping prevent premature convergence.

Answer: Fitness values are 5, 9, 13, and 17. The selected candidates are 7 and 5; this crossover gives offspring 5 and 7.

10. Training loss and validation loss

Question: During training of a neural network,th e training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance.

This pattern indicates overfitting. The network keeps improving on the training data, but its performance on validation data gets worse.

Technique 1: Regularization. Add a penalty for large weights to the loss function.

Formula: L_total = L_original + lambda sum(w_i^2)

The parameter lambda controls how strongly the penalty is applied.

Technique 2: Early stopping. Monitor validation loss and stop training when it no longer improves. Keep the model checkpoint with the lowest validation loss.

Answer: The issue is overfitting. Regularization and early stopping can help the model generalize better.

11. Perceptron model - Customer purchase prediction

Question:A perceptron model is usedto classify whethera customerwill purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Given: x1 = 2, x2 = 1, w1 = 0.5, w2 = 0.3, b = -0.4.

Formula for net input: z = w1x1 + w2x2 + b

Substitute: z = (0.5 x 2) + (0.3 x 1) - 0.4 z =

1.0 + 0.3 - 0.4 = 0.9

The step function gives 1 when z >= 0 and 0 otherwise. Since 0.9 >= 0, the output is 1.

Answer: Net input = 0.9; output = 1. The perceptron predicts that the customer makes a purchase.

12. Weight update using the delta learning rule

> ***Question:*** In a neural network, a neuron is trained using thed elta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration.

Given: w_old = 0.6, eta = 0.15, x = 2.5, y = 0.7, t = 1.

Formula for error: e = t - y

Substitute: e = 1 - 0.7 = 0.3

Formula for weight change: Delta w = eta(t - y)x

Substitute: Delta w = 0.15(1 - 0.7)(2.5) = 0.1125

Formula for new weight: w_new = w_old + Delta w

Substitute: w_new = 0.6 + 0.1125 = 0.7125

Answer: Error = 0.3, weight change = 0.1125, and new weight = 0.7125.

13. Classification model evaluation - Fraud detection

> ***Question:*** A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly.

Given: TP = 140, TN = 180, FP = 25, FN = 35. Total N = 140 + 180 + 25 + 35 = 380.

Accuracy: Formula = (TP + TN)/N x 100.

Substitute: (140 + 180)/380 x 100 = 84.21%

Precision: Formula = TP/(TP + FP) x 100.

Substitute: 140/(140 + 25) x 100 = 84.85%

Recall: Formula = TP/(TP + FN) x 100.

Substitute: 140/(140 + 35) x 100 = 80.00%

Specificity: Formula = TN/(TN + FP) x 100.

Substitute: 180/(180 + 25) x 100 = 87.80%

F1-score: Formula = 2TP/(2TP + FP + FN).

Substitute: 280/(280 + 25 + 35) = 82.35%

Metric	Result
Accuracy	84.21%
Precision	84.85%
Recall / Sensitivity	80.00%
Specificity F1-	87.80%
score	82.35%

The false-positive rate is FP/(FP + TN) x 100 = 25/205 x 100 = 12.20%. False positives can cause genuine transactions to be flagged and investigated unnecessarily. Precision is 84.85%, so some alerts will be incorrect. Whether the model is suitable depends on the cost of false alerts and missed fraud.

Answer: Accuracy = 84.21%, precision = 84.85%, recall = 80.00%, specificity = 87.80%, and F1-score = 82.35%.

14. Genetic Algorithm - Maximizing f(x) = 3x + 2

> ***Question:*** A Genetic Algorithmi s applied to maximize the functionf(x) = 3x + 2 for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.

Given: population = {2, 5, 7, 9}; fitness function f(x) = 3x + 2.

Use f(x) = 3x + 2 to calculate each fitness value.

Individual	Calculation	Fitness
2	3(2) + 2 3(5)	8
5	+ 2 3(7) + 2	17
7	3(9) + 2	23
9		29

The best two candidates are 9 and 7.

Using four-bit binary: 9 = 1001 and 7 = 0111. Choose a crossover point after the second bit.

Parents: 10|01 and 01|11. Exchange the parts after the point.

Offspring 1 = 10|11 = 1011 = 11; Offspring 2 = 01|01 = 0101 = 5

Their fitness values are f(11) = 3(11) + 2 = 35 and f(5) = 3(5) + 2 = 17.

Mutation changes one or more bits randomly. For example, 1011 can become 1010. This introduces variety and helps the algorithm explore other possible solutions.

Answer: Fitness values are 8, 17, 23, and 29. The selected candidates are 9 and 7; the illustrated offspring are 11 and 5.

15. Deep learning model - Training and validation accuracy

> ***Question:*** During thet raining of a deep learning model, thetraining accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation.

Training accuracy reaches 97%, while validation accuracy peaks at 75% and later fluctuates around 72%. This suggests overfitting.

Overfitting occurs when the model learns training examples too closely, including details that do not generalize to new data.

Technique 1: Dropout. Randomly deactivate some neurons during training so the model does not rely too heavily on particular neurons.

Technique 2: Data augmentation. Create varied training examples while preserving their labels. For images, suitable examples include small rotations, cropping, resizing, and moderate brightness changes.

Validation metrics show how the model performs on data not used to update its weights. They help identify overfitting, compare models, and select a suitable checkpoint.

Answer: The model is overfitting. Dropout and data augmentation can help improve generalization, and validation metrics show whether the model performs well beyond its training data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand